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Quality-Informed Active Learning on Social Media in Crisis Scenarios

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Abstract. During large-scale emergencies, timely situational awareness is crucial for effective response actions aimed at mitigating losses. With their real-time and distributed nature, social media provide valuable support for situational awareness. However, the unpredictability of emergency scenarios poses a challenge in understanding which data are relevant, especially in the early stages. This study explores the challenge of learning to identify relevant data dynamically as critical events unfold, operating with limited ground-truth availability in an Active Learning paradigm. We propose a tailored framework based on Information Quality indicators and evaluate it across multiple crisis events. We assess the importance of the proposed indicators for characterizing social media posts across different events, notions of relevance, and temporal intervals. We then explore the usability of these indicators to perform Active Learning in evolving emergency scenarios. Finally, we investigate whether the importance of individual posts can be generalized. Based on the insights obtained on the role of Information Quality, the steps to advance this line of research are outlined.

Keywords. Active Learning, Information Quality, Situational Awareness

1. Introduction

When large-scale crisis events occur, a timely *situational awareness* facilitates the identification of response actions. A large volume of potentially valuable data is generated, for example, on social media. However, methods for isolating *relevant* and *informative* data [1] are hindered by the non-stationary nature of data, also referred to as *concept drift* [2]. In order to update the available knowledge, understanding which data should be verified becomes then a key challenge. To mitigate the implications of concept drift, an *Active Learning* strategy (AL) can be employed to update the predictor with recent, informative samples. The aim of this paper is to explore the challenge of extracting a high-quality subset of incoming social media posts, subject to a validation budget.

Selection strategies are usually based on *information-based* features, such as uncertainty and ensemble disagreement, or *representation-based features*, such as data distribution, which describes how the density of values is allocated within their domain [3]. Information quality (IQ) indicators, which are valuable for situational awareness, are usually overlooked by AL selection strategies. In this work, we then explore the contribution of IQ indicators to characterizing data during an emergency [4]. To evaluate

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the effectiveness of IQ indicators for the AL task, we focus on the following highlights, proposing preliminary findings:

- *H1*: An analytical study of the context-dependent *indicator importance*, using selected emergency events as validation scenarios.
- *H2*: A quality-driven Active Learning method informed by the obtained results, complemented with a performance evaluation.
- *H3*: An investigation of *data item importance* for a model's performance.

In order to assess the broader applicability and robustness of the proposed method under varying conditions, the experiments are conducted on different scenarios and concepts of relevance.

2. Related work

Many efforts have been dedicated to extracting relevant data during emergency events. At the same time, emergencies are characterized by significant instability due to rapidly unfolding events. Surprisingly, only a small amount of work has focused on the task of extracting relevant data with an AL framework [5,6] or as a few-shot setting [7] in emergency scenarios, especially since adaptation to concept drift has been extensively studied [8] and the tailoring of AL methods to context has emerged as a requirement [9]. Several AL approaches try to maximize the efficiency of labelling in pool-based [10] and streaming scenarios [11]. Among the proposed approaches, *information-based* strategies leverage predictor properties, such as their uncertainty, by selecting the most uncertain items according to the current predictor [12], or their disagreement, by selecting items for which the disagreement among different models is largest [13]. In contrast, *representation-based* strategies exploit the density of the input space in an effort to select representative points [14]. A key limitation of information-based methods is that they tend to focus on points near the decision boundaries, neglecting the distribution of the data. On the other hand, representation-based approaches often need more queries to adequately cover the input space, leading to slower convergence [3]. Some authors have proposed to directly measure the expected contribution of individual data points to the performance of a predictor, for example, using estimates of Shapley values [15].

To the best of our knowledge, no investigation has been conducted on the potential of IQ indicators for the AL task. The impact of data quality and their contextual nature have been extensively studied[16]. Data quality on social media data has also been focused on [17], and related IQ indicators frameworks have been proposed for emergency scenarios [4]. However, their impact in terms of adaptation remains mainly unstudied in this scenario.

3. Methods

Definitions and quality indicators. A data item in a stream of social media items has an associated creation time t and a representation $\underline{x}$ defined over a domain $\underline{X}$. Each data item has an associated *relevance* label y that can only be discovered via annotation, which in turn is constrained by a capacity of B items per unit of time. A predictor P_T, which

outputs relevance estimates $\hat{y}$, is generated at time T, e.g., at fixed time intervals spaced by τ. P_T can be trained until data window W_T, which contains all data items for which $T - \tau \leq t < T$. The performance evaluation scheme is prequential, meaning that P_T is evaluated on data in $W_{T+\tau}$.

IQ indicators enhance the utility of data [16], which is directly related to its relevance. As W_T is processed, IQ indicators $\underline{\phi}$, defined over a domain $\underline{\Phi}$, are computed for each data item. W_T^* indicates data in W_T enriched with indicators and relevance estimates. Elaborating on the taxonomy provided in [16] and extending the framework presented in [4], the following indicators are calculated: *Text length*, *Average word length*, *Emergency keyword count*, *ContentPartsConsistency*, *Redundancy*, *Singularity*. When applicable, the proposed indicators are calculated separately for the text and the image of a post. In the remainder of the section, we define the indicators introduced in this study and refer to [4] for the remaining ones.

ContentPartsConsistency. Measures the agreement between the text and image of a post. Computed as the cosine similarity between the $\underline{x}_n.t$ embedding of a post's text and the $\underline{x}_n.i$ embedding of the LLM-derived caption of the corresponding image:

$$ContentPartsConsistency = cos(\underline{x}_n.t, \underline{x}_n.i) = \frac{\underline{x}_n.t \cdot \underline{x}_n.i}{\|\underline{x}_n.t\| \|\underline{x}_n.i\|}$$

Redundancy. Measures the presence of similar data instances. Computed as the average cosine similarity to the K nearest instances to a data item $\underline{x}_n$:

$$Redundancy = \frac{\sum_{k=1}^{K} cos(\underline{x}_n.j, \underline{x}_k.j)}{K} \quad n \neq k$$

Singularity. Measures the separateness of a data item from other items. Calculated as the minimum cosine distance of an item from any other instance:

$$Singularity = 1 - \max cos(\underline{x}_n.j, \underline{x}_k.j) \quad n \neq k$$

Emergency keyword count. To calculate the relatedness to emergency topics, we computed the number of occurrences corresponding to the CrisisLex lexicon [18].

Indicator and item importance. IQ indicators provide an abstract description of the characteristics of a data item. However, the issue of assessing the *importance* of data and their characteristics is essentially tied to the concept of relevance. Historical observations on importance, based on previous events, face limitations in their applicability and do not account for evolution. As importance is contextual, an assessment of both *indicator* and *item* importance, and their impact on AL are explored. *Indicator importance* can be evaluated using feature importance techniques in the following way. Manually labelled data is used to train a predictor using the current window of $\underline{\phi}$ indicators as a dataset, according to some loss function $\mathscr{L}(\hat{y}, y)$. SHapley Additive exPlanation (SHAP) values [19], aggregated at window level, are then used to assess the importance at indicator level. Feature importance is computed as the mean absolute SHAP value across instances, normalized using the L1 norm. The identification of data that are most valuable for learning can be mapped instead to *data item importance*. It has been proposed to apply Shapley valuation to data items[20], utilizing approximation methods due to the significantly larger combinatorial space. We use the Monte Carlo method proposed therein, truncating at empirical convergence, to estimate the importance of each data point.

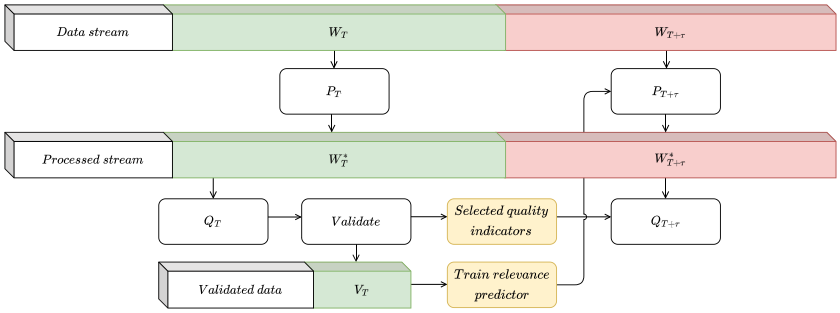

Figure 1. Diagram of the proposed AL setup

Quality-oriented Active Learning. Active Learning aims to improve the training of a predictor by selecting a subset of data items that are most suitable for the learning task. The proposed framework is detailed in Figure 1. For each input window W_T, the corresponding processed window W_T^* is computed. In order to adapt to change, P_T is replaced by an updated predictor $P_{T+\tau}$, which is trained on a subset V_T of data items that have been selected for validation from each window W_T according to a query strategy Q_T, valid at time T, and to a budget B, with the aim of maximizing the performance of $P_{T+\tau}$.

We propose to perform AL by transitioning the data representation used by Q_T to the information quality space $\underline{\Phi}$, rather than utilizing the original feature space $\underline{X}$. We hypothesize that IQ indicators are able to capture abstract properties of data, that can effectively support the AL task. We also observe that the IQ space is low-dimensional and interpretable. In Section 4, we experimentally evaluate the impact of the two proposed interpretations of importance – *indicator importance* and *item importance* – on the AL task. Regarding highlight *H2*, related to indicator importance, Q_T is implemented using Core-Set [14], which selects representative data items by minimising the largest distance measured from the remaining items, in the IQ space $\underline{\Phi}$. A variant of the query strategy Q_T, leveraging only the most important indicators measured on $W_{T-\tau}$, is also proposed. Regarding highlight *H3*, related to item importance, Q_T is implemented by regressing the importance of the items in $W_{T-\tau}$ using Random Forests [21].

4. Experiments and results

This section outlines the experimental setups used to evaluate the effect of leveraging the IQ space for gathering relevant data during emergencies. Sections 4.1 and 4.2 investigate the relevance of the proposed indicators under varying labels, events, and time (*H1*). Section 4.3 assesses AL within the IQ space and compares the approach with the literature (*H2*). Section 4.4 explores data item importance in the IQ space (*H3*).

The experiments utilize social media datasets related to different emergency events: Hurricane Maria and Sri Lanka Floods from [1]; Hurricane Harvey and Hurricane Florence from [22]; Thailand floods from [23]; and Ukrainian conflict from [24]. All the datasets are composed of Twitter posts, each with an associated image. The definition of relevance used for the labels is the *informativeness* concept taken from [1]. The characteristics of the datasets are reported in Table 1. Text embeddings are obtained using *MiniLM-L6-v2*. Image embeddings are obtained from the activations of the final layer

Event	Posts	% Relevant	Start	End
Hurricane Harvey	3635	49.9%	31/08/17	04/09/17
Hurricane Florence	1431	19.1%	14/09/17	17/09/17
Hurricane Maria	3924	63.6%	21/09/17	10/11/17
California Wildfires	1464	78.6%	12/10/17	27/10/17
Thailand Floods	1339	61.1%	26/09/21	28/09/21
Ukrainian Conflict	19657	11.9%	22/02/22	27/02/22

Table 1. Descriptions of the datasets utilized: item count, percentage of relevant items, collection timeframe

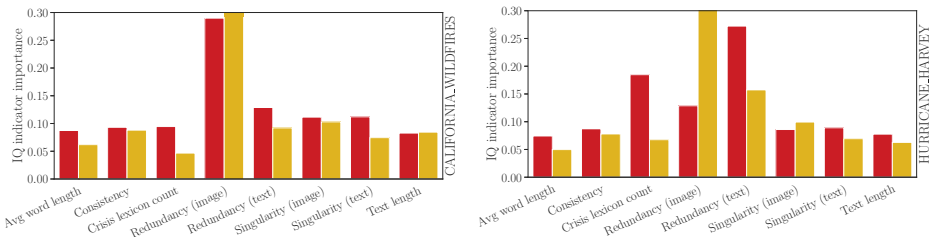

Figure 2. Indicator importance when targeting the relevance of text (*red*) or image (*yellow*) modality

of *VGG19* [25]. The image description is obtained via *LLaVA-1.5* [26] using a general-purpose prompt requesting a description of the image. Each dataset includes two relevance labels: one for the text and one for the image, both based on the *informativeness* definition provided above. In cases where a manual label was unavailable, it was generated through automatic annotation using *StableBeluga2*, averaging five runs.

4.1. Impact of context on indicator importance

The objective of the first experiment is to assess the usability of the proposed framework (*H1*). We evaluate the importance of the indicators when changing the relevance label. The experiments are repeated twice for each dataset, once for each of the text and image relevance labels. Figure 2 shows example results[2], highlighting two main aspects: (1) the most important indicators change across events (left and right panes in Figure 2); (2) the importance of the indicators within the same event changes across different concepts of relevance (red and yellow bars in Figure 2). We conclude that indicator importance is strongly context-dependent. Although case-specific, this finding suggests that IQ indicators, based on the context, can effectively be linked to relevance.

4.2. Effect of contextual dynamics on indicator importance

The second experiment, again related to highlight *H1*, is motivated by the analysis of the importance of IQ indicators over time. We evaluate the indicator importance on consecutive windows W_T of the same duration, with length τ of 8 hours, targeting the text relevance label. Figure 3 shows the results of selected experiments, highlighting two main aspects: (1) the indicator importance is highly variable over time; (2) the evolution is

[2]Only a selection is displayed due to space constraints.

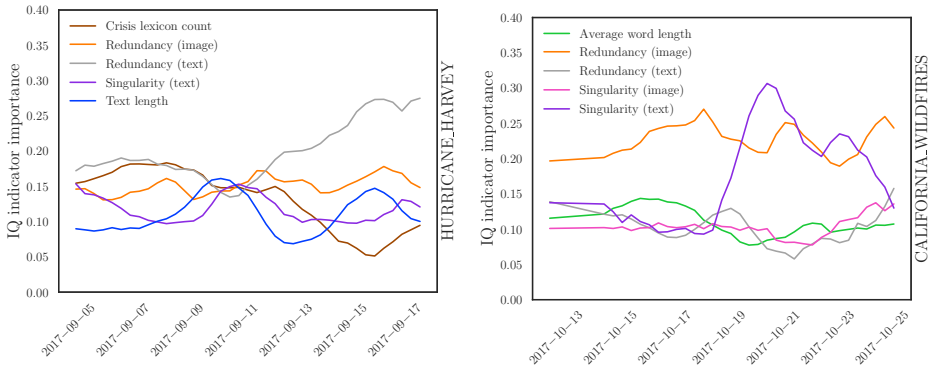

Figure 3. Temporal evolution of the top 5 feature importance scores

specific to each event. This finding is significant for AL, as it suggests that a selection strategy Q_T may benefit from utilizing current, up-to-date feature importance values. Moreover, the event specificity observed in 4.1 is confirmed.

4.3. Active Learning in the IQ space

The third experiment, related to highlight *H2*, assesses the suitability of the IQ indicators space for AL. We benchmark the method proposed in 3 varying the budget B from 10% to 100% in increments of 10%. Two variants of the method are employed:

- *coreset-iq-all*: utilizing all the IQ indicators
- *coreset-iq-best*: using the most important ones according to the previous $W_{T-\tau}$[3]

Each dataset is split into three consecutive windows of equal length, preserving the temporal ordering. The first window is reserved for warming up the uncertainty sampling predictor and the indicator performances for the *coreset-iq-best* variant of the proposed method. The selection Q is performed on the second window, and a predictor P is trained on the selected data in the original feature space. P is implemented as a fully connected feedforward neural network. Then, prequential evaluation is performed using all the data in the third window, using the macro-averaged F1 score. We perform the experiments on different datasets, comparing the proposed methods with the following baselines: random sampling, Core-Set [14], and uncertainty sampling [12], which involves selecting the B elements for which the prediction confidence, as given by P_T, is lowest.

Figure 4 shows the results of the experiment for selected datasets, highlighting: (1) the robustness of the predictive accuracy when transitioning from the original feature space $\underline{X}$ to the IQ space $\underline{\Phi}$; (2) how the selection of the most relevant indicators seems preferable in all scenarios. The proposed method shows an improvement in performance with respect to the baselines for B values of 0.1 and 0.2. Although derived from a limited set of scenarios, these results suggest that performing AL in the IQ space $\underline{\Phi}$ preserves essential information for AL performance (see also the coreset performance for comparison). This is significant because the dimensionality of $\underline{\Phi}$ is significantly lower.

We formulated two hypotheses to account for this result. On the one hand, relative measures tend to be more robust to some transformations in the data distributions (e.g.,

[3]Indicators accounting for 75% of the overall importance, 5 features on average

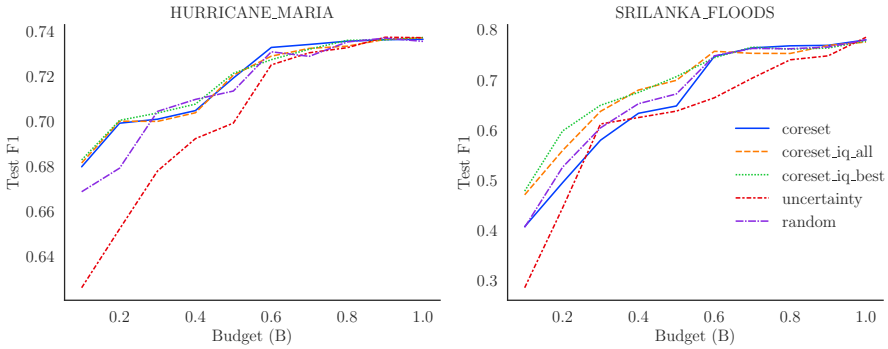

Figure 4. F1 performance over different budgets and datasets

isometries and translations), providing measures that could generalize as the data distribution changes. As an example, the *Redundancy* and *Singularity* indicators measure relative distances in the original feature space. On the other hand, introducing quality indicators that are correlated with relevance offers additional information to the sampling strategy and possibly helps to obtain label-balanced data points, thereby enhancing the learning process. For instance, in the proposed scenarios the *Frequency of emergency keywords* is usually positively correlated with relevant posts.

4.4. Item importance in the IQ space

In line with highlight *H3*, we investigate whether directly estimating the importance of data items is suitable for AL purposes. The calculation of the Shapley value of each item φ_i is framed as an expectation calculation problem, according to [20]:

$$\varphi_i = \mathbb{E}_{\pi \sim \Pi} \left[V\left(S_\pi^i \cup \{i\}\right) - V\left(S_\pi^i\right) \right]$$

where Π is the uniform distribution over the $n!$ possible permutations of data items and S_π^i is the set of data items preceding item i in permutation π. The calculation is performed until convergence. Given that $\underline{\Phi}$ is low-dimensional, the Monte Carlo estimation can be performed efficiently, as the base step of the calculation requires less time. As reported in [15], we regress the importance values of the data points using Random Forest [21] on their IQ representation $\underline{\phi}_i$, calculated on W_T, and assess the predictions on $W_{T+\tau}$. We observe that, while the approach is able to generalize the importance of items across partitions of the same window W_T, it fails to predict the importance of future data items in $W_{T+\tau}$, indicating that the approach as-is is not suitable for AL in emergency scenarios, possibly due to the temporal distribution of the datasets, which span across many days.

To test whether the results depend on the IQ space or the regression model, we conduct an additional experiment. A sliding window W_T of width $\tau = 24h$ is shifted by smaller steps of width $\lambda = 1h$ with overlaps, and item importance is tested at each step on a future test window ($8h$ after W_T), to evaluate its temporal consistency. We monitor important items from the last $4h$ of the first window. Figure 5 shows selected results obtained using different days of the last dataset in Table 1. The solid line represents the median importance of the monitored items within three percentiles of the green region – 20, 50, and 100 – while the dashed line shows the same statistics but calculated across the entire W_T windows. As a first observation, items initially selected as important tend

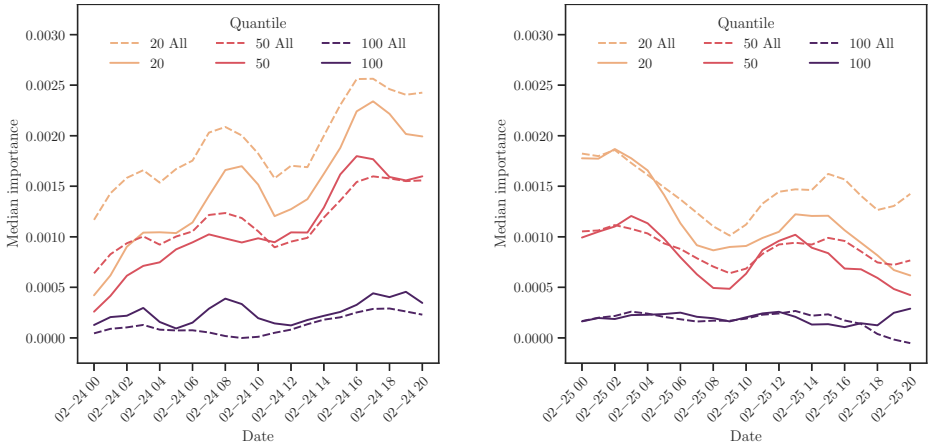

Figure 5. Median importance of the reference (solid line) and full (dashed line) item set, over time

to maintain their importance over time. Moreover, the importance of these items evolves differently, growing or decaying over time depending on the context (Figure 5 left and right panes show opposite trends). This result suggests that importance estimates may be valuable since they are consistent over time.

5. Conclusions and future work

This study highlights exploratory but encouraging results on the use of information quality indicators for analyzing social media data during emergencies. These indicators capture salient characteristics of data, contributing to the individuation of relevant items and ultimately enhancing context awareness. Notably, indicator importances are time-sensitive and context-bound, underscoring the importance of their dynamic evaluation as events unfold. We show that the IQ space can effectively support Active Learning (AL) with consistent behaviour and low budgets. We also show that the importance of data items is consistent over time, even if further research is needed to fully harness this information, possibly contributing to the quantification of the evolution dynamic, which could be linked to the AL task, for example, by parameterizing the selection strategy Q_T.

A current limitation of this study is a systematic exploration of fine-grained windowing approaches, which will be addressed in future work. The same holds for the absence of a streaming-based setup, which would better reflect real-world use cases and facilitate meaningful comparisons. Moreover, the results are limited to the proposed set of indicators and scenarios. The method utilized in Section 4.4 does not scale well to larger datasets, highlighting the need for further exploration of heuristics aimed at data-intensive scenarios.

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